

A Packing of 15 Four-Spaces in $\mathbb{R}^{12}$

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Abstract

We give 15 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 2.856822. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.844075, obtained by deleting 1 of the 16 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 15 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.856822\dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 15$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.844075, obtained by deleting 1 of the 16 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 15/14 = 2.857143$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N15.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 65 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
28568228172216963026373088592921297624067584079908055026397894615946852
248233445252714640727587089095472054648168045900653986305233671018369291
4965852966042878975344067876815241305858314941108
```

denominator:

```
99999999999162714171836501485868395616468118107452920608086039947156752
660021788280899483841821293657218831311973543577029390270870869530447240
422666442472954374602748476928463756517960716407
```

B Coordinates

15 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N15.txt`.

```
-414649310727 253933106182 102026850951 2191615256853684438530 190229307874 -365682441975 -468025022871625136539 426019651707 137988475126 -524136636738
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